

Digitalization—for Port Traffic Management

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Abstract—The purpose of this project is to use a Decision Support System (DSS) to optimise pre-gate parking management at small EU ports. Budgetary restrictions, maximising the service area, minimising the time spent looking for a parking space, maximising traffic flow, COVID-19 safety precautions, seasonal traffic fluctuations, environmental compliance, high utilisation rates, and future flexibility are all covered by the DSS. The database allows for quick access to information such as ferry times and parking arrangements. Queuing theory, layout optimisation, and predictive analysis are some of the models used in the DSS. A DBMS, machine learning, heuristics, data analytics, and simulation software are used in implementation to improve efficiency, safety, and regulatory compliance at these ports.

I. INTRODUCTION

The increasing demand for efficient and safe port operations has driven the need for digitalization in port traffic management, particularly in small European ports handling Roll-on/Roll-off (Ro-Ro) operations. These ports play a critical role in facilitating the transportation of trucks, trailers, and cars across various routes in Europe. As these ports expand their operations to accommodate new berths and terminals, the management of pre-gate parking areas becomes a crucial aspect of overall port efficiency and safety.

This project aims to explore the digitalization of port traffic management with a specific focus on optimizing pre-gate parking layouts. By leveraging modern technologies and data-driven approaches, the objective is to provide decision support for various parking layout scenarios based on quantified metrics such as budget calculations, service area capacities, and average slot availability times. Additionally, the project seeks to analyze layout performance indicators using queuing theory to predict queue lengths, the number of check-in gates required, and the per-amount of traffic or defined queue sizes.

The report is structured to provide a comprehensive analysis of the current challenges in port traffic management, the scope of digitalization efforts, and the methodologies employed to gather and process data. It also discusses the design and implementation of optimization models, the integration of these models into existing port systems, and the expected impact on

decision-making processes. Through a small-scale experiment, the effectiveness of proposed solutions will be tested and validated, providing actionable insights and recommendations for port authorities and stakeholders involved in port traffic management.

II. DIVISION OF LABOUR

This team of researchers has made significant contributions that have been extremely helpful in the development of this decision assistance system. Team members engaged in a range of tasks, including documentation, programming, research, and more. The following table presents the total amount of contributions made to the decision support system:

Lawrence E. Henesey: Project Supervisor

TABLE I: Division of Work

S.NO	NAME	WORK DIVISION
1	AHMED MAHFOOZ ALI KHAN	A,B,C
2	TAFAZZUL HAZQUEEL SYED	C,D,F
3	SATISH RAJ SOMAROUTHU	B,D,F
4	KOSARI SUHAS	E,G

The distribution of work among the team members based on tasks is displayed in Table [?]. The following tasks were finished for the project:

- A - Manager
- B - Designer
- C - Documenter of the report
- D - Logger (team session events)
- E - Presenter
- F - Viewer/Reviewer
- G - Reviser (Project Revisions)

Task that are in project:

- 1) Ferry arrival time, departing time, gates open time.
- 2) Geometry and size of parking area
- 3) Entry point and exit gates
- 4) Service traffic inside parking area

III. PROBLEM DESCRIPTION

A. Problem specification

This involves figuring out both the initial building costs of the parking facility and the ongoing maintenance expenses to make sure the layout makes financial sense. It looks at how essential service areas like payment machines, electric vehicle charging stations, and other amenities are spread out within the parking facility. It also counts the total number of parking spaces in different layout scenarios, including spaces for different vehicle types and those with special needs. Additionally, it estimates the average time it takes for a driver to find an available spot, considering both peak and off-peak times, which impacts user satisfaction and the efficiency of the facility. Finally, it examines how easy it is for users to navigate the parking facility and reach key areas, focusing on how close parking spaces are to exits and entrances, and the overall user experience.

This indicator measures the likelihood of queues forming at various points within the parking facility, helping to identify potential bottlenecks. It determines the optimal number of check-in gates needed to handle expected traffic volumes, ensuring smooth entry and exit processes. It evaluates the expected length of queues during peak hours, providing insights into congestion levels and aiding in designing layouts that minimize wait times. This metric also measures the average time it takes for vehicles to be processed at check-in gates, which influences the overall efficiency of the parking facility.

Analyzing lengths of queues and wait times at entry gates when ferries arrive at their busiest times aids in fine-tuning gate operations and enhancing traffic flow. Evaluating how parking spaces are used compared to the total capacity of the parking area gives us a grasp of parking demand and spots where congestion might occur. By calculating the average time it takes for a vehicle to enter, park, and leave the parking area, we can gauge the effectiveness of parking operations and pinpoint areas that could be made more efficient.

B. Stakeholders

Port Authorities: These are government or corporate entities responsible for managing and operating ports. They oversee activities such as vessel traffic management, cargo handling, security, and infrastructure maintenance within the port area. Their primary goal is to ensure efficient and safe maritime operations while generating revenue for the port and surrounding communities.

Shipping Companies: These are businesses that own or operate ships for transporting cargo and/or passengers by sea. They manage the logistics of moving goods from one port to another, including scheduling shipments, arranging for loading and unloading of cargo, and ensuring compliance with international maritime regulations. Shipping companies play a crucial role in global trade by facilitating the movement of goods between countries and regions.

Ferry Drivers: Ferry drivers are responsible for operating ferries, which are boats or ships used to transport passengers,

vehicles, and cargo across bodies of water, such as rivers or straits. They ensure the safe navigation of the vessel, adhere to maritime regulations, maintain communication with port authorities, and provide assistance to passengers as needed.

Passengers: Passengers are individuals who travel by sea for various reasons, including tourism, business, or personal matters. They may use passenger ships, ferries, or cruise liners to reach their destinations. Passengers rely on shipping companies and port authorities to provide safe and comfortable travel experiences, including amenities, entertainment, and accommodation during their journey. **Question of interest**

Port Authorities: Port authorities oversee the bustling hub of maritime activity, and their primary question of interest revolves around maintaining the smooth flow of ships in and out of the port. They're tasked with ensuring that vessels can navigate safely through the port's waters, docking and departing efficiently while adhering to stringent safety regulations. They ponder over questions like: How can we optimize our infrastructure to handle increasing traffic? What measures can we implement to mitigate environmental impacts? How do we balance security needs with the facilitation of trade?

Shipping Companies: For shipping companies, the question of interest is all about maximizing efficiency and profitability. They're constantly seeking ways to streamline their operations, minimize costs, and optimize shipping routes. Their inquiries delve into topics such as fuel consumption reduction strategies, cargo handling techniques to minimize turnaround times, advanced route planning algorithms to avoid congested areas or adverse weather conditions, and compliance with international shipping regulations to avoid penalties and delays.

Ferry Drivers: Ferry drivers, the captains and crews responsible for navigating vessels through waterways, are primarily concerned with ensuring the safe and efficient transport of passengers and cargo. Their central question revolves around how to navigate their vessels effectively while prioritizing safety above all else. They ponder over questions like: What are the current weather and water conditions, and how might they affect our journey? Are there any navigational hazards we need to be aware of? How can we efficiently load and unload passengers and cargo while maintaining stability and safety?

Passengers: Passengers, the lifeblood of ferry services, are primarily interested in the quality of their travel experience. Their central question revolves around how to ensure a comfortable and reliable journey from point A to point B. They seek answers to questions such as: What are the available departure times and ticketing options? What amenities are offered onboard, and how comfortable are the seating arrangements? Can they trust that the ferry service will run on time and provide clear communication in case of delays or disruption? Overall, passengers want assurance that their journey will be smooth, safe, and pleasant.

IV. DESCRIPTION AND MOTIVATION OF SCOPE

Description of Scope: The focus of this project is on the digitalization of port traffic management, with a specific emphasis on pre-gate parking areas at small EU ports that handle Roll-on/Roll-off (Ro-Ro) operations. These ports accommodate a variety of vehicles, including trucks, trailers, and passenger cars. The goal is to develop and implement advanced decision support systems (DSS) to optimize parking layouts, manage traffic flow, and ensure safety and operational efficiency. By concentrating on pre-gate parking, the project aims to create a foundational framework that can be expanded to other areas of port management in the future.

The project involves a thorough analysis and redesign of pre-gate parking layouts, utilizing digital tools and technologies. This includes the implementation of specialized check-in gates for different vehicle types, real-time monitoring systems, and simulation models to predict and manage traffic behavior. By integrating these elements, the project aims to develop a seamless and efficient parking management system that can adapt to varying traffic conditions and demands. The scope also includes the development of performance indicators based on queuing theory to evaluate the effectiveness of different layouts and enable data-driven decision-making.

A key aspect of the project is stakeholder engagement and data management. The project will involve collaboration with various stakeholders, including port authorities, shipping companies, truck drivers, and passengers, to gather comprehensive data and insights. This data will be essential in designing and refining the DSS, ensuring it meets the practical needs and challenges faced by different users. Additionally, the project will establish protocols for data collection, processing, and interpretation, ensuring that the system is both reliable and scalable.

Motivation: The motivation for this project arises from the urgent need to enhance the efficiency and safety of port operations amidst growing traffic volumes and increasing operational complexities. As ports expand to include new berths and terminals, efficient management of parking and traffic flows becomes critical. Traditional methods are no longer sufficient to handle the scale and complexity of modern port operations. Digitalization, through the use of decision support systems (DSS), offers a robust solution by providing real-time data analysis, predictive modeling, and automated decision-making capabilities. This not only improves operational efficiency but also reduces congestion, enhances safety, and provides a better experience for all stakeholders involved.

Addressing the inefficiencies and safety concerns associated with current pre-gate parking management practices is a significant motivator for this project. Challenges such as prolonged wait times, congestion, and increased risk of accidents highlight the need for a more effective approach. By leveraging digital solutions, the project aims to mitigate these issues, ensuring smooth and orderly traffic flow within the port. The integration of real-time data and advanced analytics will enable port authorities to make informed decisions, optimize

resource allocation, and respond promptly to any incidents or changes in traffic patterns. This proactive approach not only enhances operational efficiency but also aligns with broader goals of sustainability and technological advancement in port management.

Engaging stakeholders and managing data effectively are crucial for the success of this project. The motivation behind this lies in developing a system that is user-centric and grounded in real-world requirements. By involving stakeholders in the design and implementation process, the project ensures that the solutions developed are practical, feasible, and widely accepted. Effective data management is also essential for the accuracy and reliability of the DSS. By establishing robust data protocols, the project aims to provide port authorities with the tools needed to continuously monitor and improve their operations, driving long-term benefits and sustainable growth.

V. INPUT DATA MANAGEMENT

A. Data Gathering and Processing

1) Data Sources:

- **Ferry Operations Data:** This dataset encompasses detailed information about ferry operations, including the types and quantities of trucks and cars transported by each ferry. It also records the specific arrival and departure times of ferries, as well as the times at which the gates open for boarding and disembarking.
- **Parking Area Data:** This dataset includes comprehensive details about the parking area, such as its geometry and dimensions, the locations of entry and exit points, and the number of check-in gates available.
- **Service Traffic Data:** This dataset contains information on the various types of service vehicles operating within the parking area, along with the frequency of their movements.
- **Safety and Regulatory Requirements:** This dataset provides information on safety and regulatory protocols, including COVID-19 guidelines, measures to prevent incidents, and strategies for managing congestion.

2) Data Processing:

- **Data Collection:** Automated data collection systems for real-time tracking of vehicles.
- **Data Integration:** Combining data from various sources into a centralized database using Python scripts.
- **Data Cleaning:** Using Python libraries like Pandas to remove duplicates, correct errors, and ensure data consistency.
- **Data Analysis:** Using statistical tools and libraries in Python to analyze traffic patterns, peak times, and average dwell times.

VI. SOLUTION ARCHITECTURE

A. Code Snippets

API Code

```
from flask import Flask, request, jsonify, render_template
import pandas as pd
import numpy as np

app = Flask(__name__)

# Load the data with error handling
file_path = '/Users/ahmedmahfoozalikhana/Downloads/Port data.xlsx'
datetime_format = '%H:%M:%S' # Example format, adjust as needed

try:
    port_data = pd.read_excel(file_path, sheet_name='Sheet1')
    # Convert time columns to datetime
    for col in port_data.columns[1:]:
        port_data[col] = pd.to_datetime(port_data[col], format=datetime_format, errors='coerce').dt.time
except FileNotFoundError:
    port_data = pd.DataFrame()
    print("File not found.")
except Exception as e:
    port_data = pd.DataFrame()
    print(f"An error occurred: {e}")

@app.route('/', methods=['GET'])
def home():
    return render_template('home.html')

@app.route('/submit-form', methods=['POST'])
def submit_form():
    data = {
        'arrival_time': request.form.get('arrival_time'),
        'departing_time': request.form.get('departing_time'),
        'gates_open_time': request.form.get('gates_open_time'),
        'parking_geometry': request.form.get('parking_geometry'),
        'entry_exit_gates': request.form.get('entry_exit_gates'),
        'service_traffic': request.form.get('service_traffic')
    }
    return jsonify(data), 200

@app.route('/transit-times/average', methods=['GET'])
def get_average_transit_times():
    if port_data.empty:
        return jsonify({'error': 'Data not available'}), 500

    transit_times = []
    for index, row in port_data.iterrows():
        try:
```

```
            depart_time = pd.to_datetime(str(row['Unnamed: 1']), format=datetime_format)
            arrive_time = pd.to_datetime(str(row['Unnamed: 2']), format=datetime_format)
            transit_time = (arrive_time - depart_time).total_seconds() / 60 # in minutes
            transit_times.append(transit_time)
        except Exception:
            continue
    avg_transit_time = np.mean(transit_times)
    if transit_times else None
    return jsonify({'average_transit_time': avg_transit_time})

@app.route('/transit-times/actual', methods=['GET'])
def get_actual_transit_times():
    if port_data.empty:
        return jsonify({'error': 'Data not available'}), 500

    actual_transit_times = []
    for index, row in port_data.iterrows():
        try:
            depart_time = pd.to_datetime(str(row['Unnamed: 1']), format=datetime_format)
            arrive_time = pd.to_datetime(str(row['Unnamed: 2']), format=datetime_format)
            transit_time = (arrive_time - depart_time).total_seconds() / 60 # in minutes
            actual_transit_times.append({'depart_time': str(row['Unnamed: 1']), 'arrive_time': str(row['Unnamed: 2']), 'transit_time': transit_time})
        except Exception:
            continue
    return jsonify({'actual_transit_times': actual_transit_times})

@app.route('/waiting-times/average', methods=['GET'])
def get_average_waiting_times():
    if port_data.empty:
        return jsonify({'error': 'Data not available'}), 500

    waiting_times = []
    for index, row in port_data.iterrows():
        try:
            depart_time = pd.to_datetime(str(row['Unnamed: 1']), format=datetime_format)
            arrive_time = pd.to_datetime(str(row['Unnamed: 2']), format=datetime_format)
            waiting_time = (arrive_time - depart_time).total_seconds() / 60 # in minutes
```



```

<div class="form-group">
  <label for="gates_open_time">
    Gates Open Time</label>
  <input type="text" class="form
    -control" id="
      gates_open_time" name="
      gates_open_time"
      placeholder="HH:MM:SS">
  <small class="form-text text-
    muted">Example: 06:00:00</
    small>
</div>
<div class="form-group">
  <label for="parking_geometry">
    Parking Geometry</label>
  <input type="text" class="form
    -control" id="
      parking_geometry" name="
      parking_geometry"
      placeholder="e.g., 200x100
        meters, 50 slots for
        trucks, 100 slots for cars
        ">
  <small class="form-text text-
    muted">Example: 200x100
        meters, 50 slots for
        trucks, 100 slots for cars
        </small>
</div>
<div class="form-group">
  <label for="entry_exit_gates">
    Entry/Exit Gates</label>
  <input type="text" class="form
    -control" id="
      entry_exit_gates" name="
      entry_exit_gates"
      placeholder="Entry/Exit
        Gate Details">
  <small class="form-text text-
    muted">Example: 3 entry
        gates, 2 exit gates</small
        >
</div>
<div class="form-group">
  <label for="service_traffic">
    Service Traffic</label>
  <input type="text" class="form
    -control" id="
      service_traffic" name="
      service_traffic"
      placeholder="Service
        Traffic Details">
  <small class="form-text text-
    muted">Example: Peak hours
        from 08:00 to 10:00</
        small>
</div>
<button type="submit" class="btn
  btn-primary">Submit</button>
</form>
</div>
<script src="https://code.jquery.com/
  jquery-3.5.1.slim.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/
  @popperjs/core@2.5.4/dist/umd/popper.
  min.js"></script>
<script src="https://stackpath.

```

```

bootstrapcdn.com/bootstrap/4.5.2/js/
bootstrap.min.js"></script>
</body>
</html>

```

VII. DEFINITION AND MOTIVATION OF CHOICES/ALTERNATIVES

A. Choices/Alternatives

1) Layout Scenarios:

- **Predefined Parking Configurations:** These are predetermined parking arrangements derived from historical data analysis.
- **Dynamic Layouts:** These are adaptive parking configurations that adjust in real-time according to current traffic conditions.

2) Decision Variables:

- **Number of Parking Slots:** Indicates the total count of slots available for both trucks and cars.
- **Service Areas:** Specifies the quantity and positions of service areas within the facility.
- **Check-in Gates:** Identifies the number and placements of check-in gates, categorized by vehicle types.
- **Safety Zones:** Highlights designated areas intended for safety measures and emergency responses.

B. Representation in the System

- **Finite Set of Choices:** A limited number of pre-designed layout scenarios.

Dynamic Layout: **Description:** Adaptive configuration that changes based on real-time traffic conditions.

- **ferry Parking Slots:** 20-30(variable)
- **Service Areas:** 4-6 (variable)
- **Check-in Gates for Ferry :** 4-5 (variable)
- **Safety Zones:** 4-6 (variable)

- **Optimization Model:** Combination of decision variables used in optimization algorithms to generate efficient layouts.

Data for Optimization Model

Objective Function:

- Minimize the average wait time for vehicles to find a free slot.
- Maximize the utilization of parking slots.

Constraints:

- Total parking slots (FPS) must not exceed the parking area capacity.
- Service areas must be adequately distributed to ensure minimal interference with parking slots.
- Safety zones must cover at least 10% of the total parking area.

Optimization Variables:

- **FPS** (ferry Parking Slots): 20 to 130
- **NSA** (Number of Service Areas): 4 to 6
- **SAS** (Service Area Size): 100 to 250 square meters
- **CIGT** (Check-in Gates for ferry): 4 to 5
- **NSZ** (Number of Safety Zones): 3 to 4
- **SZA** (Safety Zone Area): 50 to 70 square meters

VIII. MODEL DESIGN/BUILD

A. System Design

1) Subsystems:

- **Data Management Subsystem:** Handles data collection, storage, and processing using Python.
- **Optimization Subsystem:** Implements optimization algorithms in Python to generate and evaluate layout scenarios.
- **Simulation Subsystem:** Uses simulation tools in Python to simulate the performance of different layouts under various conditions.
- **User Interface Subsystem:** Provides an interactive interface using HTML, CSS, Python and Flask for users to input data, view results, and make decisions.

2) Technologies Used:

IX. APPROPRIATE USE OF THE SYSTEM

A. Output Data Interpretation

- **Capacity Utilization:** Visualization of how efficiently the parking area is used.
- **Average Wait Times:** Graphical representation of average time to find a free slot and wait times at check-in gates.
- **Queue Length Probability:** Charts showing the probability of queues of different lengths.
- **Safety Compliance:** Indicators for adherence to safety protocols and congestion management.

B. Presentation of Output Data

- **Dashboards:** Interactive dashboards with real-time data and historical trends.
- **Reports:** Detailed reports with metrics and recommendations.
- **Visualizations:** Heat maps, flow diagrams, and other visual aids to illustrate traffic patterns and layout efficiency.
- **Python:** For data processing, optimization algorithms, and simulation.
- **Flask:** To create the web application and serve the user interface.
- **HTML/CSS:** To design the user interface, ensuring it is user-friendly and responsive.

X. MODEL IMPACT AND DECISION-MAKING

A. Impact on Decision Making

- **Efficiency Improvements:** Recommendations for layouts that maximize capacity and minimize wait times.

- **Safety Enhancements:** Identifying configurations that ensure compliance with safety protocols.
- **Cost-Effectiveness:** Budget analysis to balance cost and operational efficiency.

B. Decision-Making Process

- **Scenario Comparison:** Decision-makers can compare different layout scenarios based on key performance indicators.
- **What-if Analysis:** Assessing the impact of changes in traffic volume, ferry schedules, or safety requirements.
- **Continuous Improvement:** Using feedback and performance data to iteratively improve parking layouts.

XI. SMALL-SCALE EXPERIMENT

A. Experiment Description

Objective: Test the effectiveness of the DSS in a controlled environment.

1) Setup:

- **Simulated Parking Area:** A scaled-down model of a port parking area.
- **Traffic Simulation:** Use Python-based simulation tools to emulate vehicle movement and congestion.
- **Data Collection:** Real-time data collection on vehicle arrival, parking, and departure times using Python scripts.

2) Procedure:

- **Baseline Scenario:** Implement a basic parking layout without optimization.
- **Optimized Scenario:** Apply the DSS to design and implement an optimized layout.
- **Performance Metrics:** Measure and compare key metrics (e.g., wait times) for both scenarios.

XII. REFERENCES

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Fig. 1: API of Ferry Port Data

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